U0-Anchor Multiple-Level Batch Gallery

March 31, 2026

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Overview

This note collects the multiple-level summary plots for the six benchmark cases run with the paired schedule

$$N = [2, 4, 6, 8, 10, 12, 14, 18], \qquad p = [3, 5, 7, 9, 11, 13, 15, 19],$$

with target continuation values $t = 10^{-p}$ at each level.

Each case section contains:

- the summary plot `u0_anchor_multiple_level_error_vs_N.png`,
- the per-level continuous-norm table extracted from the corresponding CSV summary file.

1 Batch Summary

Table 1: Batch summary for the six multiple-level benchmark cases.

Case	Levels	Success	Final N	Final p	Final residual	Final ContL2	Final ContH1	CPU(s)
`mild_exp_separable`	8	true	18	19	5.760e-20	5.261e-15	5.621e-13	32.756
`mixed_polynomial_convex`	8	true	18	19	5.551e-17	1.028e-14	1.122e-12	37.761
`mild_sextic_convex`	8	true	18	19	1.041e-16	1.042e-14	1.136e-12	31.608
`gaussian_bump_convex`	8	true	18	19	6.047e-16	5.177e-13	2.062e-11	38.346
`quartic_convex`	8	true	18	19	1.221e-15	2.597e-14	2.890e-12	30.603
`scaled_quartic_convex`	8	true	18	19	2.776e-17	5.160e-15	5.786e-13	31.359

2 mild_exp_separable

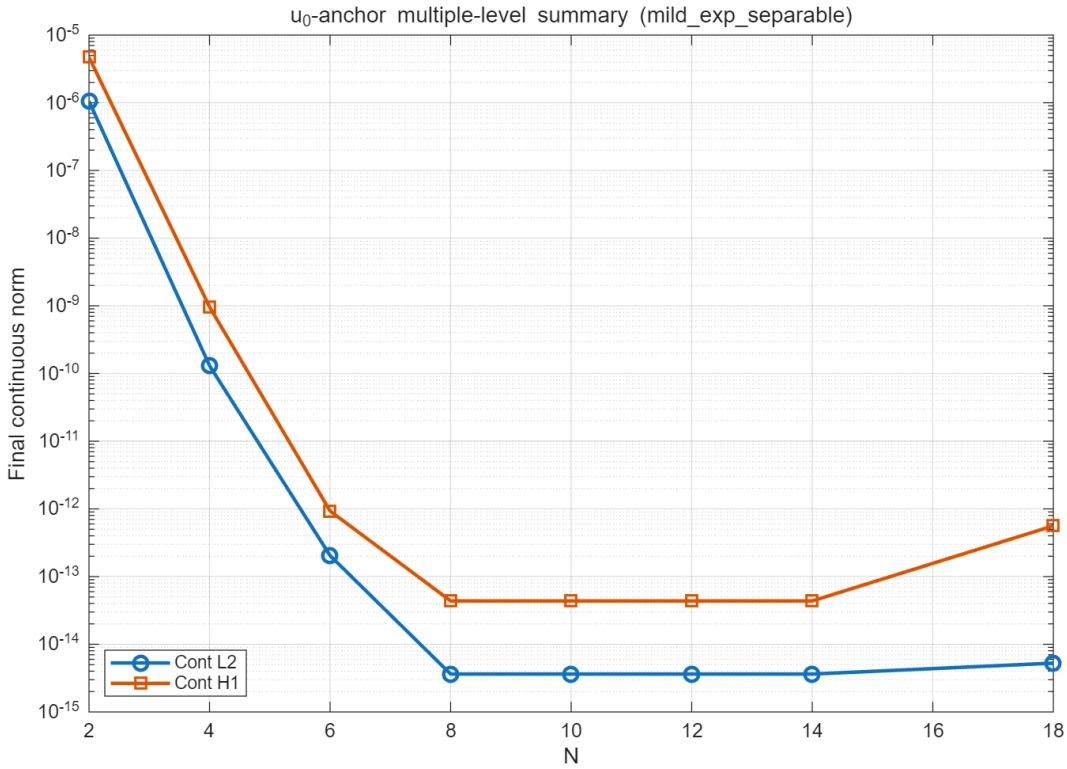

Figure 1: Multiple-level summary plot for `mild_exp_separable`.

Table 2: Per-level summary for `mild_exp_separable`.

Level	N	p	start t	target t	ContL2	ContH1	CPU(s)
1	2	3	1e+00	1e-03	1.055e-06	4.719e-06	4.144
2	4	5	1e-03	1e-05	1.305e-10	9.638e-10	4.070
3	6	7	1e-05	1e-07	2.042e-13	9.304e-13	4.059
4	8	9	1e-07	1e-09	3.629e-15	4.367e-14	3.913
5	10	11	1e-09	1e-11	3.629e-15	4.367e-14	3.761
6	12	13	1e-11	1e-13	3.629e-15	4.367e-14	4.367
7	14	15	1e-13	1e-15	3.629e-15	4.367e-14	4.118
8	18	19	1e-15	1e-19	5.261e-15	5.621e-13	4.323

3 mixed_polynomial_convex

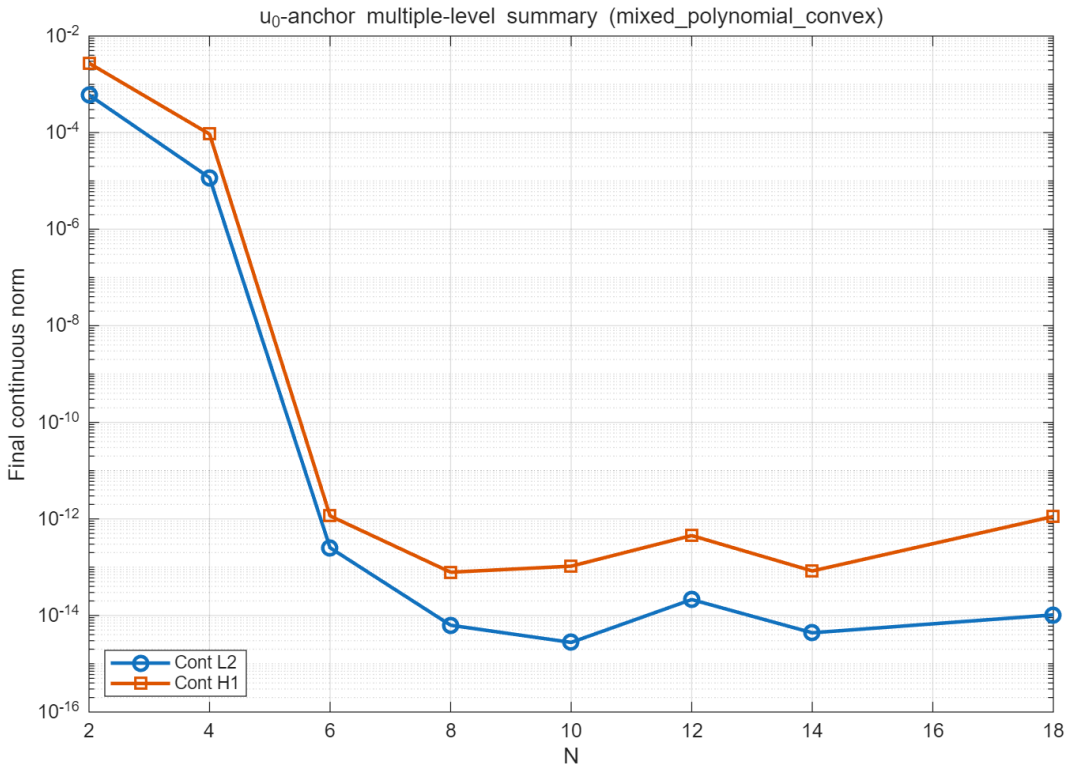

Figure 2: Multiple-level summary plot for `mixed_polynomial_convex`.

Table 3: Per-level summary for `mixed_polynomial_convex`.

Level	N	p	start t	target t	ContL2	ContH1	CPU(s)
1	2	3	1e+00	1e-03	6.025e-04	2.743e-03	4.300
2	4	5	1e-03	1e-05	1.164e-05	9.326e-05	4.072
3	6	7	1e-05	1e-07	2.522e-13	1.152e-12	4.186
4	8	9	1e-07	1e-09	6.271e-15	7.856e-14	4.279
5	10	11	1e-09	1e-11	2.767e-15	1.049e-13	3.934
6	12	13	1e-11	1e-13	2.136e-14	4.519e-13	3.886
7	14	15	1e-13	1e-15	4.377e-15	8.350e-14	5.305
8	18	19	1e-15	1e-19	1.028e-14	1.122e-12	7.800

4 mild_sextic_convex

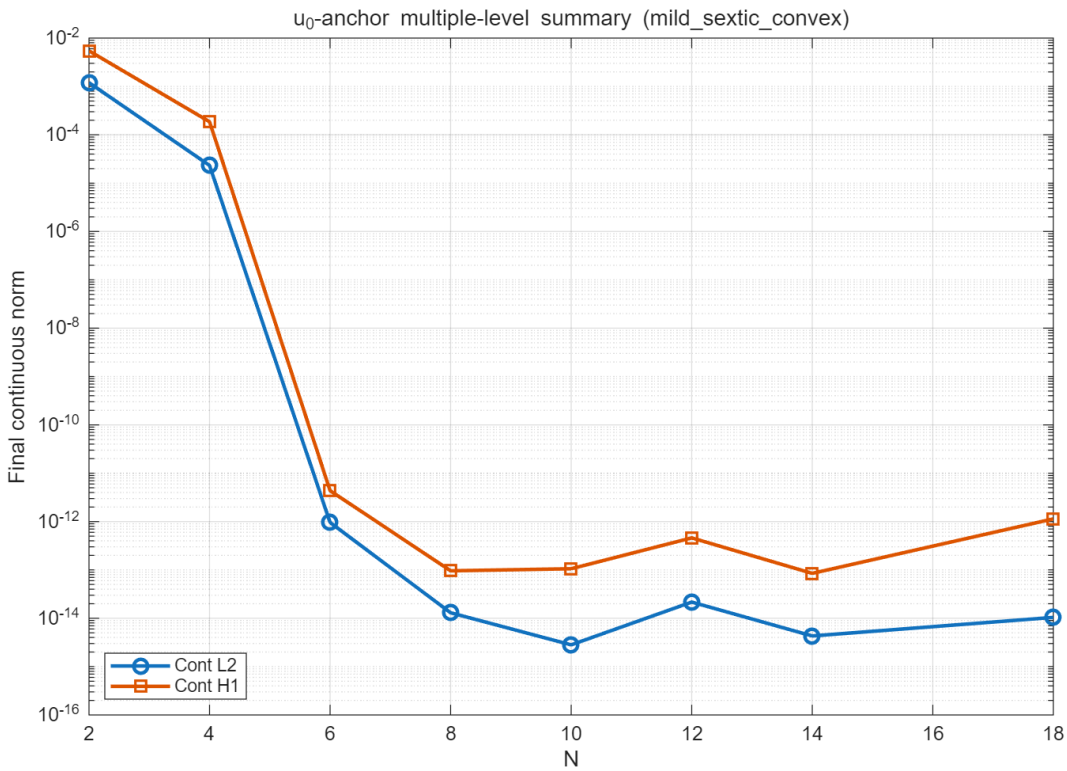

Figure 3: Multiple-level summary plot for `mild_sextic_convex`.

Table 4: Per-level summary for `mild_sextic_convex`.

Level	N	p	start t	target t	ContL2	ContH1	CPU(s)
1	2	3	1e+00	1e-03	1.197e-03	5.450e-03	3.795
2	4	5	1e-03	1e-05	2.322e-05	1.863e-04	3.722
3	6	7	1e-05	1e-07	9.583e-13	4.356e-12	3.604
4	8	9	1e-07	1e-09	1.299e-14	9.582e-14	3.817
5	10	11	1e-09	1e-11	2.819e-15	1.057e-13	3.908
6	12	13	1e-11	1e-13	2.162e-14	4.594e-13	4.061
7	14	15	1e-13	1e-15	4.309e-15	8.486e-14	4.477
8	18	19	1e-15	1e-19	1.042e-14	1.136e-12	4.224

5 gaussian_bump_convex

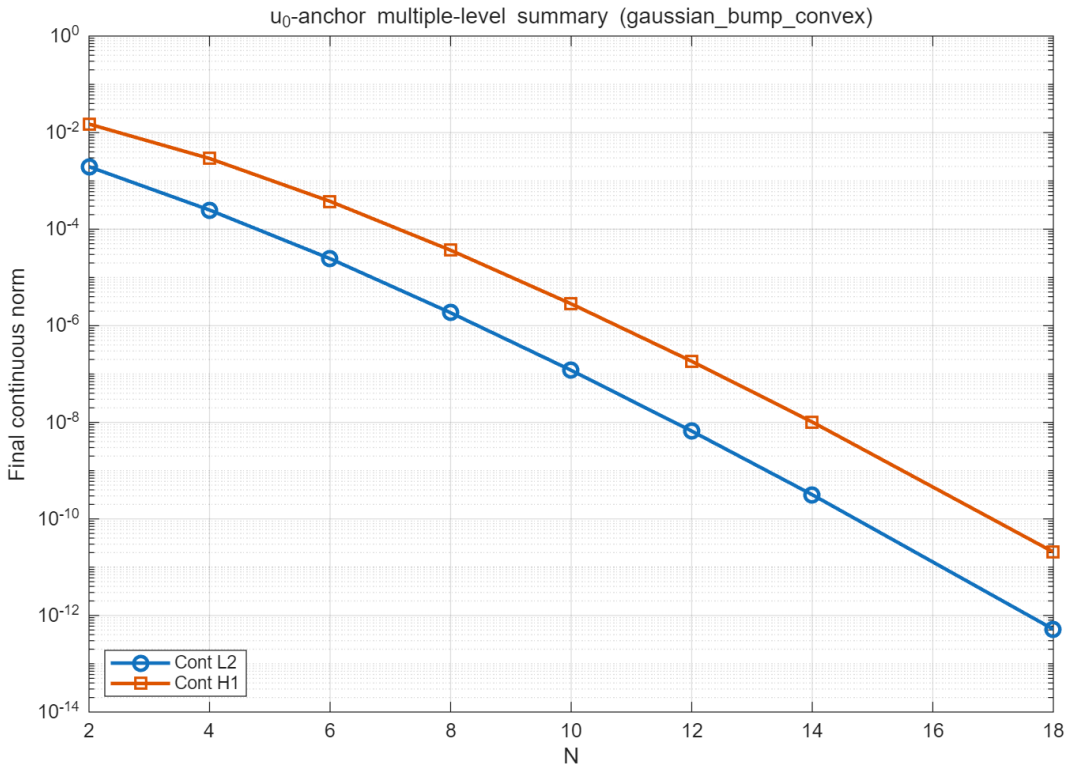

Figure 4: Multiple-level summary plot for `gaussian_bump_convex`.

Table 5: Per-level summary for `gaussian_bump_convex`.

Level	N	p	start t	target t	ContL2	ContH1	CPU(s)
1	2	3	1e+00	1e-03	1.967e-03	1.505e-02	4.756
2	4	5	1e-03	1e-05	2.492e-04	2.910e-03	4.064
3	6	7	1e-05	1e-07	2.452e-05	3.771e-04	3.823
4	8	9	1e-07	1e-09	1.860e-06	3.665e-05	3.796
5	10	11	1e-09	1e-11	1.196e-07	2.846e-06	10.226
6	12	13	1e-11	1e-13	6.579e-09	1.838e-07	3.763
7	14	15	1e-13	1e-15	3.168e-10	1.016e-08	3.692
8	18	19	1e-15	1e-19	5.177e-13	2.062e-11	4.225

6 quartic_convex

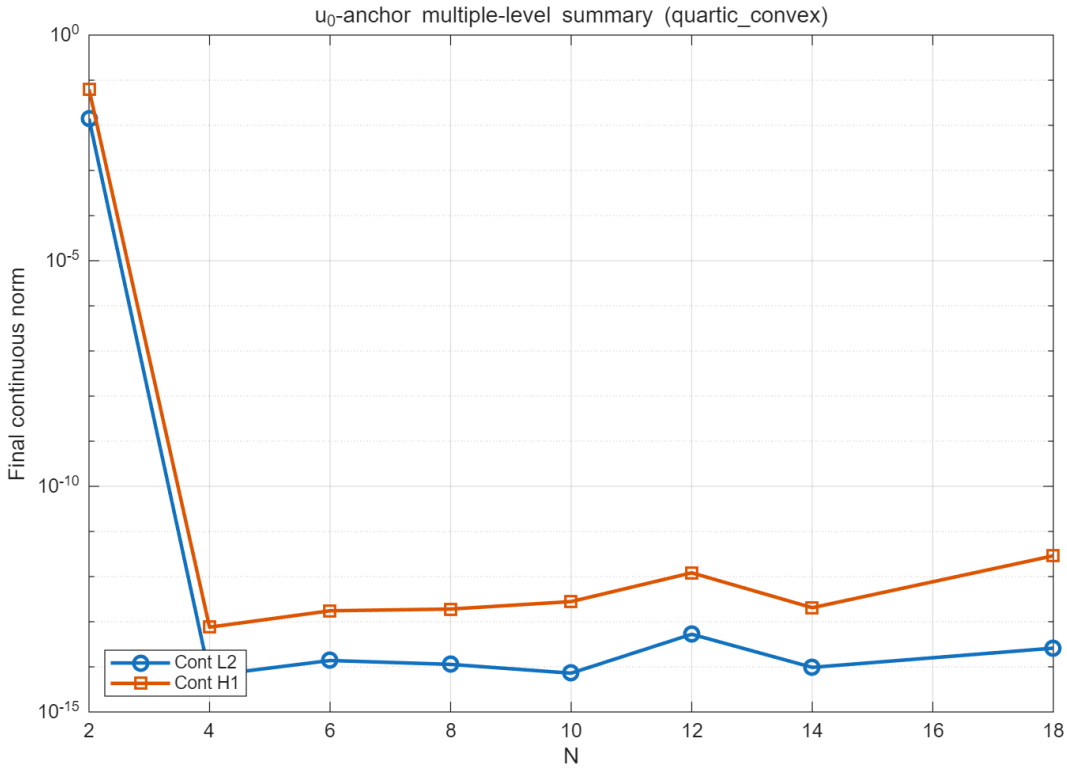

Figure 5: Multiple-level summary plot for `quartic_convex`.

Table 6: Per-level summary for `quartic_convex`.

Level	N	p	start t	target t	ContL2	ContH1	CPU(s)
1	2	3	1e+00	1e-03	1.403e-02	6.309e-02	3.628
2	4	5	1e-03	1e-05	6.416e-15	7.569e-14	3.570
3	6	7	1e-05	1e-07	1.382e-14	1.747e-13	4.012
4	8	9	1e-07	1e-09	1.134e-14	1.908e-13	3.968
5	10	11	1e-09	1e-11	7.239e-15	2.794e-13	4.320
6	12	13	1e-11	1e-13	5.277e-14	1.213e-12	3.732
7	14	15	1e-13	1e-15	9.733e-15	2.042e-13	3.715
8	18	19	1e-15	1e-19	2.597e-14	2.890e-12	3.658

7 scaled_quartic_convex

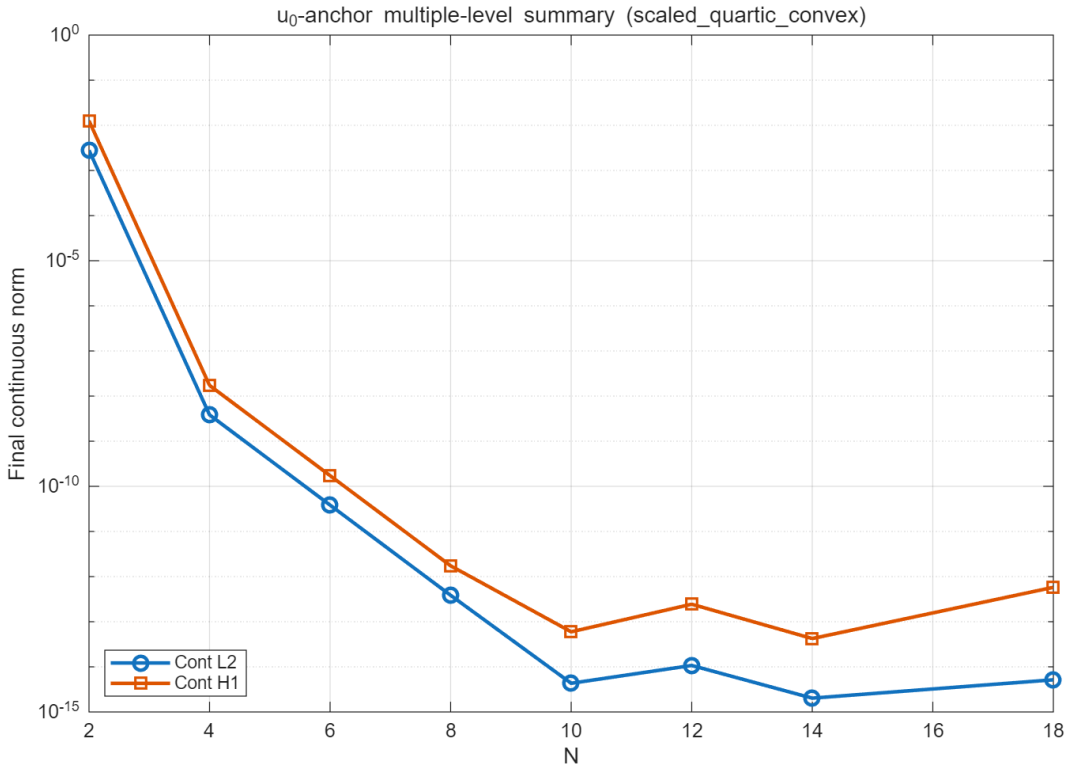

Figure 6: Multiple-level summary plot for `scaled_quartic_convex`.

Table 7: Per-level summary for `scaled_quartic_convex`.

Level	N	p	start t	target t	ContL2	ContH1	CPU(s)
1	2	3	1e+00	1e-03	2.807e-03	1.262e-02	3.790
2	4	5	1e-03	1e-05	3.835e-09	1.727e-08	4.073
3	6	7	1e-05	1e-07	3.835e-11	1.727e-10	3.862
4	8	9	1e-07	1e-09	3.849e-13	1.734e-12	3.965
5	10	11	1e-09	1e-11	4.324e-15	5.976e-14	3.909
6	12	13	1e-11	1e-13	1.073e-14	2.442e-13	3.782
7	14	15	1e-13	1e-15	2.025e-15	4.232e-14	3.837
8	18	19	1e-15	1e-19	5.160e-15	5.786e-13	4.141